

STEP 1: Install & Import Libraries

```
!pip install nltk pandas

Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex<=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.1)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)

import pandas as pd
import nltk
import re
from collections import defaultdict, Counter

nltk.download('punkt')
nltk.download('averaged_perceptron_tagger')

[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data] /root/nltk_data...
[nltk_data] Unzipping taggers/averaged_perceptron_tagger.zip.
True
```

STEP 2: Load your Uploaded Dataset

```
file_path = '/content/Twitter_Data.csv'
df = pd.read_csv(file_path)
print(df.head())

      clean_text  category
0  when modi promised "minimum government maximum...  -1.0
1  talk all the nonsense and continue all the dra...    0.0
2  what did just say vote for modi welcome bjp t...    1.0
3  asking his supporters prefix chowkidar their n...    1.0
4  answer who among these the most powerful world...    1.0

import pandas as pd

# STEP-2: Load the uploaded Twitter dataset
file_path = "/content/Twitter_Data.csv"
df = pd.read_csv(file_path)

# Verify dataset
print(df.shape)
df.head()
```

(162980, 2)

	clean_text	category
0	when modi promised "minimum government maximum...	-1.0
1	talk all the nonsense and continue all the dra...	0.0
2	what did just say vote for modi welcome bjp t...	1.0
3	asking his supporters prefix chowkidar their n...	1.0
4	answer who among these the most powerful world...	1.0

STEP-3: Preprocess Tweet Text

```
import re

def preprocess_tweet(text):
    text = str(text)
    text = re.sub(r"http\S+", "", text) # remove URLs
    text = re.sub(r"@w+", "", text) # remove mentions
    text = text.lower().strip()
    return text

df['processed_text'] = df['clean_text'].apply(preprocess_tweet)

# Check result
```

```
df[['clean_text', 'processed_text']].head()
```

	clean_text	processed_text
0	when modi promised "minimum government maximum...	when modi promised "minimum government maximum...
1	talk all the nonsense and continue all the dra...	talk all the nonsense and continue all the dra...
2	what did just say vote for modi welcome bjp t...	what did just say vote for modi welcome bjp t...
3	asking his supporters prefix chowkidar their n...	asking his supporters prefix chowkidar their n...
4	answer who among these the most powerful world...	answer who among these the most powerful world...

step 4:tokenization

```
import nltk
nltk.download('punkt_tab')

df['tokens'] = df['processed_text'].apply(nltk.word_tokenize)

df[['processed_text', 'tokens']].head()
```

```
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
```

	processed_text	tokens
0	when modi promised "minimum government maximum...	[when, modi, promised, ", minimum, government,...
1	talk all the nonsense and continue all the dra...	[talk, all, the, nonsense, and, continue, all,...
2	what did just say vote for modi welcome bjp t...	[what, did, just, say, vote, for, modi, welcom...
3	asking his supporters prefix chowkidar their n...	[asking, his, supporters, prefix, chowkidar, t...
4	answer who among these the most powerful world...	[answer, who, among, these, the, most, powerfu...

STEP-5: POS Tagging using NLTK (Weak Supervision)

```
nltk.download('averaged_perceptron_tagger_eng')
df['pos_tags'] = df['tokens'].apply(nltk.pos_tag)
df[['tokens', 'pos_tags']].head()
```

```
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]   /root/nltk_data...
[nltk_data]   Package averaged_perceptron_tagger_eng is already up-to-
[nltk_data]   date!
```

	tokens	pos_tags
0	[when, modi, promised, ", minimum, government,...	[(when, WRB), (modi, NN), (promised, VBD), (" ,...
1	[talk, all, the, nonsense, and, continue, all,...	[(talk, NN), (all, PDT), (the, DT), (nonsense,...
2	[what, did, just, say, vote, for, modi, welcom...	[(what, WP), (did, VBD), (just, RB), (say, VB)...
3	[asking, his, supporters, prefix, chowkidar, t...	[(asking, VBG), (his, PRP\$), (supporters, NNS)...
4	[answer, who, among, these, the, most, powerfu...	[(answer, NN), (who, WP), (among, IN), (these,...

STEP-6: Initialize HMM Counters

```
from collections import defaultdict, Counter

transition_counts = defaultdict(Counter)
emission_counts = defaultdict(Counter)
tag_counts = Counter()
```

STEP-7: Build HMM Counts

```
for sentence in df['pos_tags']:
    prev = "<START>"
    for word, tag in sentence:
        transition_counts[prev][tag] += 1
        emission_counts[tag][word] += 1
        tag_counts[tag] += 1
        prev = tag
    transition_counts[prev]["<END>"] += 1
```

STEP-8: Transition Probabilities

```
transition_probs = {
    prev: {tag: c/sum(tags.values()) for tag, c in tags.items()}
    for prev, tags in transition_counts.items()
}
```

```
transition_probs["DT"]

{'JJ': 0.19611021972634488,
 'NN': 0.5027632927753265,
 'DT': 0.013003075277443508,
 'RBS': 0.00421179302045728,
 'VBZ': 0.01014507287070464,
 'JJR': 0.003420689040424299,
 'NNS': 0.11655412933992958,
 'CD': 0.014245442795382627,
 '<END>': 0.011014173017783126,
 'WP': 0.00941525159335027,
 'RB': 0.01664103935463743,
 'JJS': 0.01808953959976824,
 'VBP': 0.015777510362347907,
 'CC': 0.004975041226545438,
 'IN': 0.010462628693675625,
 'VBD': 0.010022507465347417,
 'PRP$': 0.005643579801221197,
 'VBG': 0.008022462896109106,
 'PRP': 0.004830191202032357,
 'MD': 0.00811160137273254,
 'VBN': 0.005560012479386727,
 'VB': 0.001816196461202478,
 'WDT': 0.0012200828987832598,
 'WRB': 0.003955519900164906,
 'RBR': 0.0010752328742701788,
 'EX': 0.00021170388198065695,
 'NNP': 0.0012702232918839417,
 'FW': 0.0005961135624192183,
 'PDT': 0.00014485002451308107,
 'NNPS': 0.00021170388198065695,
 'UH': 1.671346436689397e-05,
 'RP': 0.00042897891875027854,
 'WP$': 3.342692873378794e-05}
```

STEP-9: Emission Probabilities

```
emission_probs = {
    tag: {w: c/sum(words.values()) for w, c in words.items()}
    for tag, words in emission_counts.items()
}

emission_probs["NN"]
```

```
'bcoz': 0.00023869365792973709,  
'mms': 0.00029735565860738436,  
'incompetency': 1.314837946223128e-05,  
'blatant': 1.921686229095341e-05,  
'misuse': 9.507289764998002e-05,  
'misgovernance': 1.7194034681379366e-05,  
'misconduct': 1.04665534044000e-05
```

STEP-10: Rare Words Count

```
rare_words = [w for tag in emission_counts for w in emission_counts[tag] if emission_counts[tag][w] == 1]  
  
len(rare_words)
```

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STEP-11: Viterbi

```
test_tweet = ["this", "movie", "lit"]  
tags = list(tag_counts.keys())  
  
def t(p, c): return transition_probs.get(p, {}).get(c, 1e-6)  
def e(tag, w): return emission_probs.get(tag, {}).get(w, 1e-6)  
  
v = {tag: t("<START>", tag) * e(tag, test_tweet[0]) for tag in tags}  
  
for word in test_tweet[1:]:  
    v = {  
        curr: max(v[prev] * t(prev, curr) * e(curr, word) for prev in tags)  
        for curr in tags  
    }  
  
max(v, key=v.get)
```

'NN'

```
best_tag = max(v, key=v.get)  
print("Final predicted POS tag:", best_tag)
```

Final predicted POS tag: NN